



Youssef Hadjkacem

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*Final-year computer engineering student at ENSI, specialized in **applied AI** (NLP, LLMs, LoRA fine-tuning, multi-agent RAG). Deployed an AI copilot to production on AWS. Seeking a 4-to-6-month ML/AI Engineering internship.*

Education

National School of Computer Science (ENSI) <i>Engineering Degree – Computer Science (Final Year)</i>	Sept. 2024 – 2027 Tunis, Tunisia
Faculty of Sciences of Sfax <i>Preparatory Classes – Mathematics (Rank: 110/1800)</i>	June 2022 – June 2024 Sfax, Tunisia

Professional Experience

AI Developer Intern, Odoo <i>June 2026 – August 2026</i>	Yonnov'IA Remote, France
– Sales teams lacked a reliable tool to generate quotes ; AI copilot combining sales history, CRM data and a sector Knowledge Graph (NetworkX), integrated into Odoo and exposed via FastAPI REST API on AWS, with an MCP interface for external agents. – Technologies: Python, FastAPI, NetworkX, Ollama, AWS, MCP	
AI Intern, Biomechanics & Exoskeleton <i>June 2026 – August 2026</i>	University of Lorraine Remote, France
– A reinforcement-learning exoskeleton controller needs a faithful biomechanical model ; 3D digital twin (Blender) from monocular motion capture, musculoskeletal analysis in OpenSim, preparing MuJoCo/MyoSuite integration via isolated Docker setups. – Technologies: Python, Blender, OpenSim, MuJoCo, MyoSuite, Docker	
Software Engineering Intern <i>June 2025 – August 2025</i>	Fogits Sfax, Tunisia
– Manually entering LinkedIn leads was slow and error-prone ; Python scraper (Selenium + BeautifulSoup) processing 800+ profiles, loaded via Docker into an Odoo ERP (-85% manual entry). – Technologies: Python, Selenium, Docker, Odoo	

Projects

GenAI Pipeline & Industrial RAG System

- Reliably answering business questions requires real sources ; full **RAG system (FAISS, vector database)** with few-shot **prompt engineering** and **LLM-as-judge** evaluation, orchestrated by an **autonomous AI agent** (LangGraph, ReAct) on a local LLM.
- **Technologies:** LangChain, LangGraph, FAISS, Django REST

AI Emergency Savior

- Recommending an emergency care provider requires balancing real-time constraints ; **LoRA fine-tuning** of **Whisper** for **voice symptom** extraction (ICD-10/SNOMED CT mapping) and **multi-objective optimization (NSGA-II)**, demo deployed on Hugging Face Spaces.
- **Technologies:** Whisper, LoRA, NSGA-II, FastAPI, TypeScript

LLM Fine-tuning for Ticket Classification

- Automatically classifying support ticket intent requires adapting a general-purpose model to a business domain ; **fine-tuning** an open-source LLM via **PEFT**, with **MLflow** experiment tracking and **quantization** (GGUF) for local inference.
- **Technologies:** Hugging Face, PEFT, MLflow, Quantization

Technical Skills

Languages: Python, C, C++, Java	Deep Learning: PyTorch, TensorFlow, Neural Networks, CNN
NLP / LLMs: LangChain, LangGraph, Ollama, LoRA	Vector Search: FAISS, embeddings, Vector Databases
Backend: FastAPI, Django REST, Docker, Odoo/CRM	Cloud: AWS (EC2, S3, IAM)
MLOps: MLflow, Distributed Training, Model Evaluation	Tools: Git, Linux, CUDA/GPU

Languages

Arabic: Native	English: C1	French: B2
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Leadership & Volunteering

ENSI Junior Enterprise (EJE) – International Relations Officer	Tunis, Tunisia
AIESEC – Python Teacher	Târgu Jiu, Romania
AIESEC – English Teacher	Gaziantep, Turkey